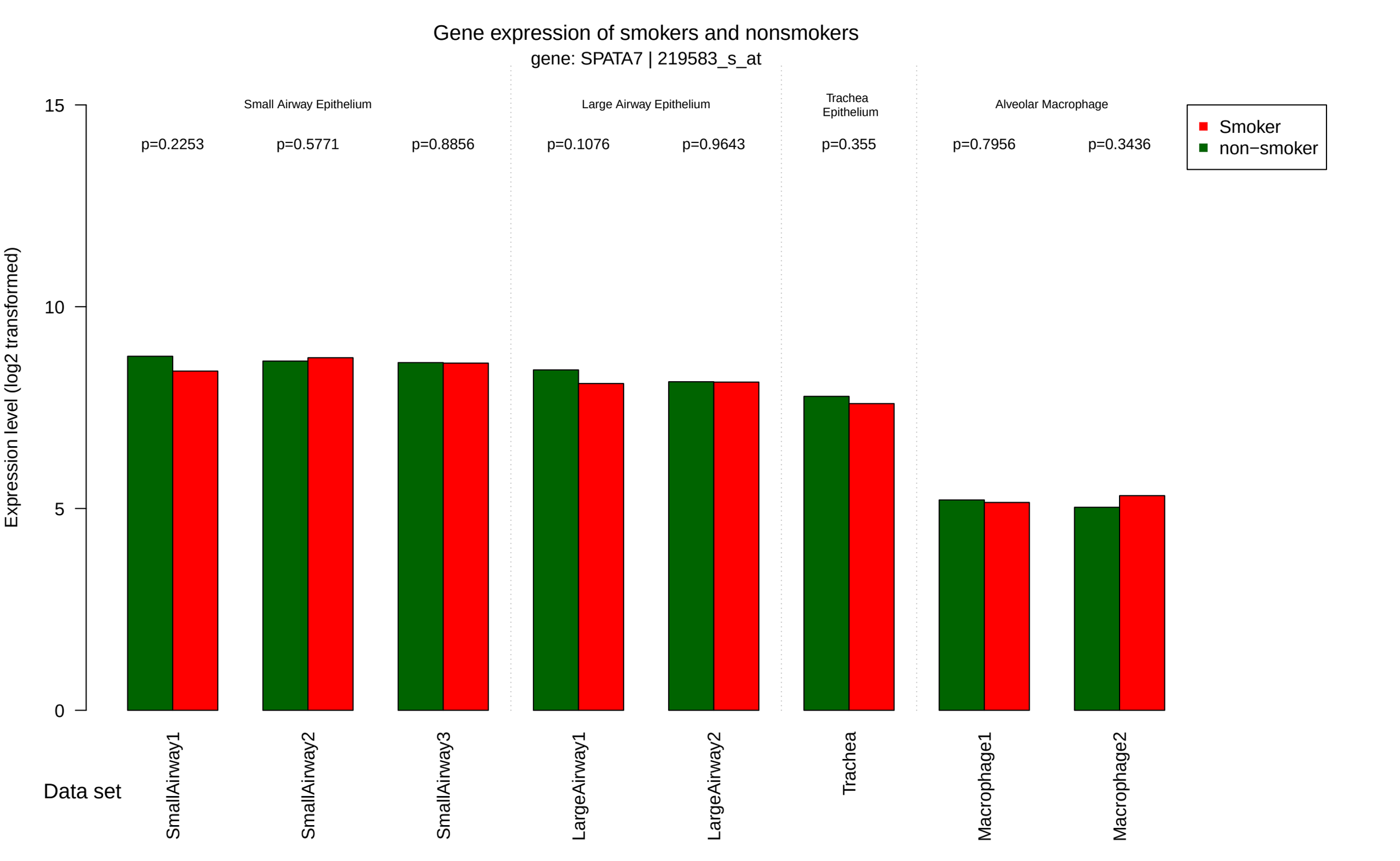
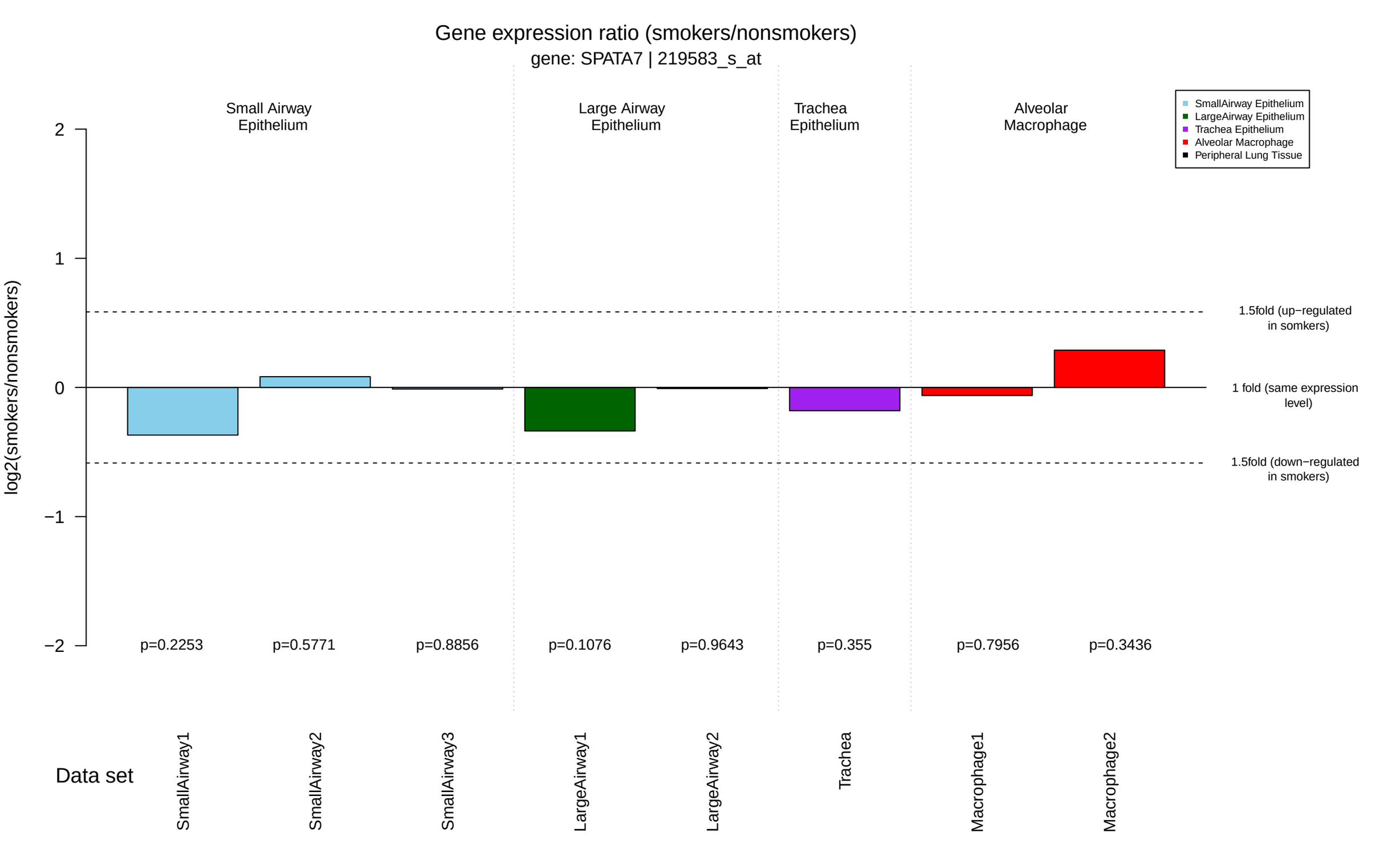
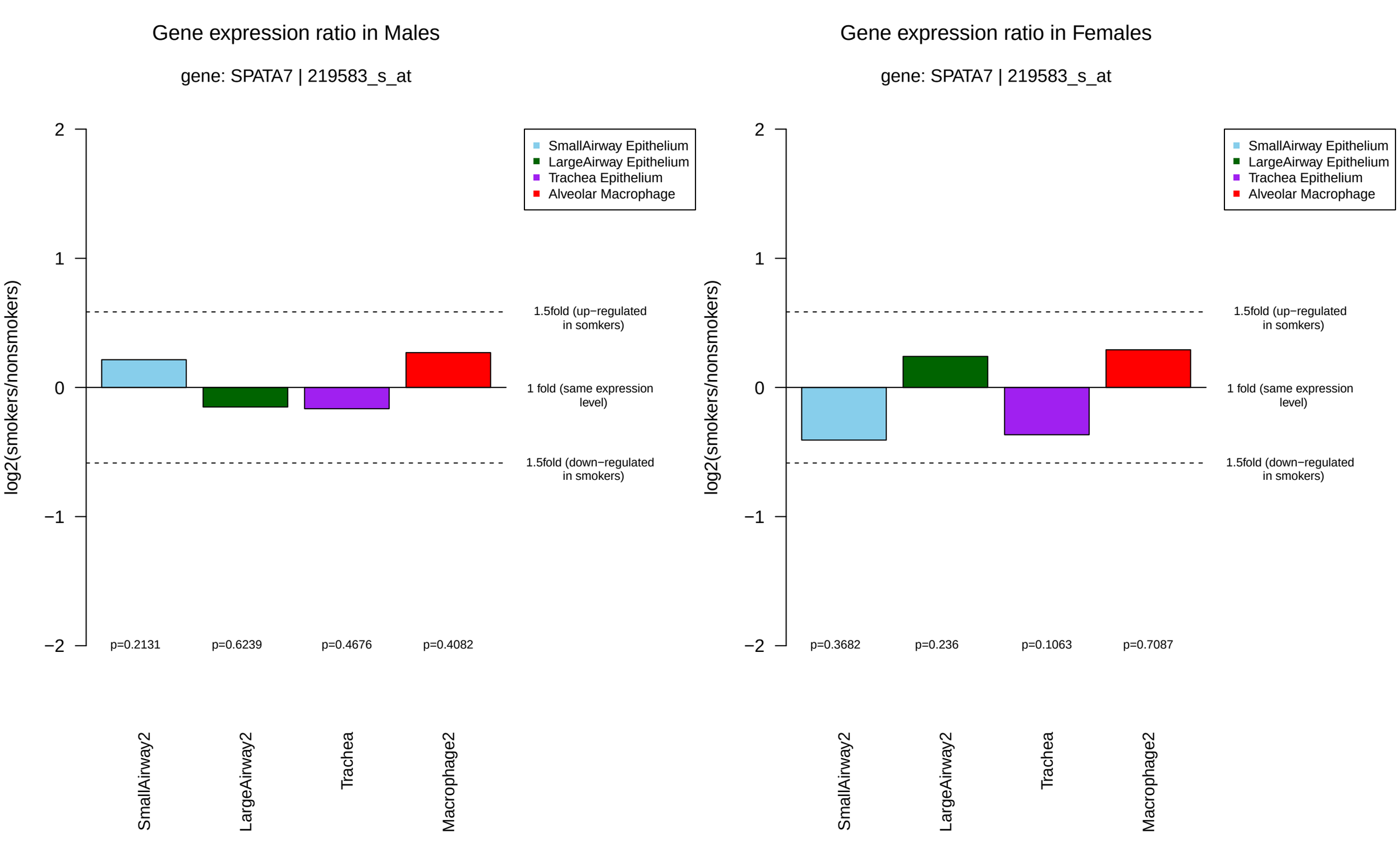
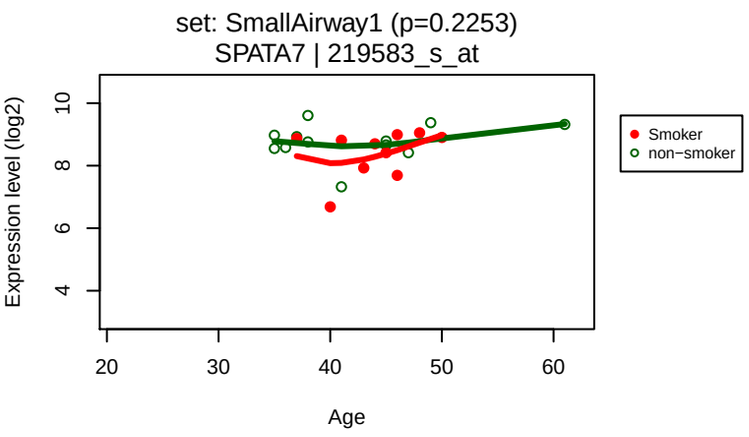
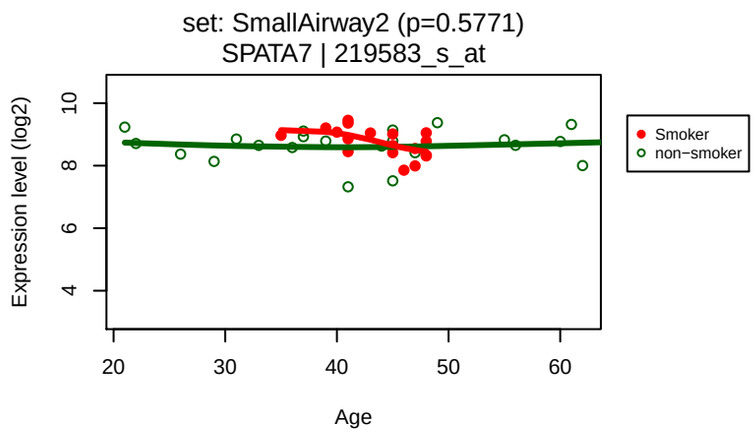
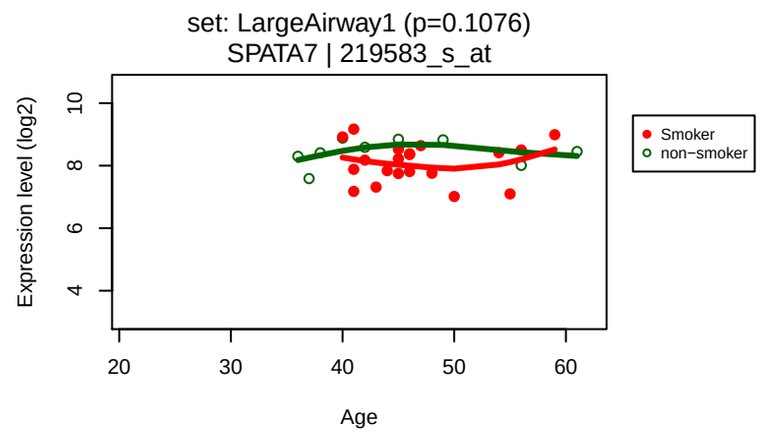

Gene expression of smokers and nonsmokers
gene: SPATA7 | 219583_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2253
p=0.5771
p=0.8856
p=0.1076
p=0.9643
p=0.355
p=0.7956
p=0.3436
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

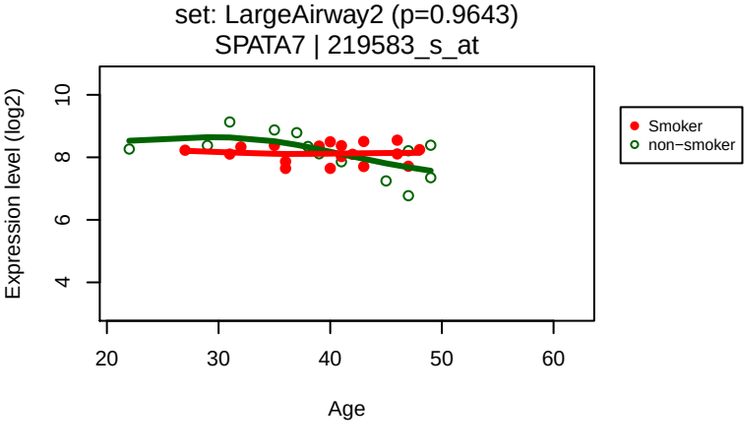

Gene expression ratio (smokers/nonsmokers)
gene: SPATA7 | 219583_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2253
p=0.5771
p=0.8856
p=0.1076
p=0.9643
p=0.355
p=0.7956
p=0.3436
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

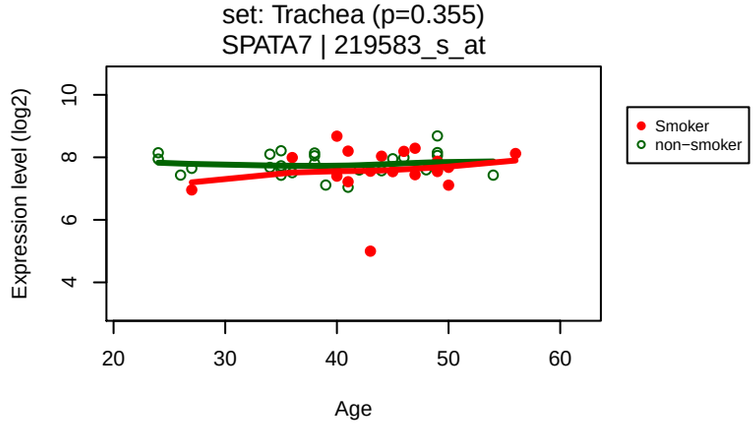

Gene expression ratio in Males
gene: SPATA7 | 219583_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2131
p=0.6239
p=0.4676
p=0.4082
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SPATA7 | 219583_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3682
p=0.236
p=0.1063
p=0.7087
SmallAirway2
LargeAirway2
Trachea
Macrophage2

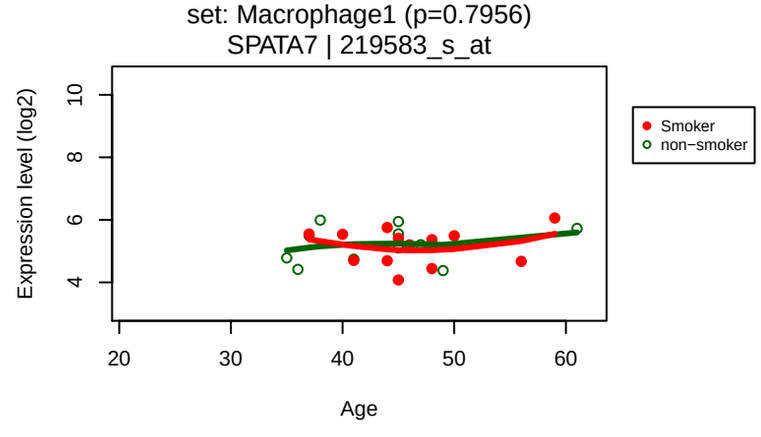

set: SmallAirway1 (p=0.2253)
SPATA7 | 219583_s_at
Expression level (log2)
Age
Smoker
non−smoker

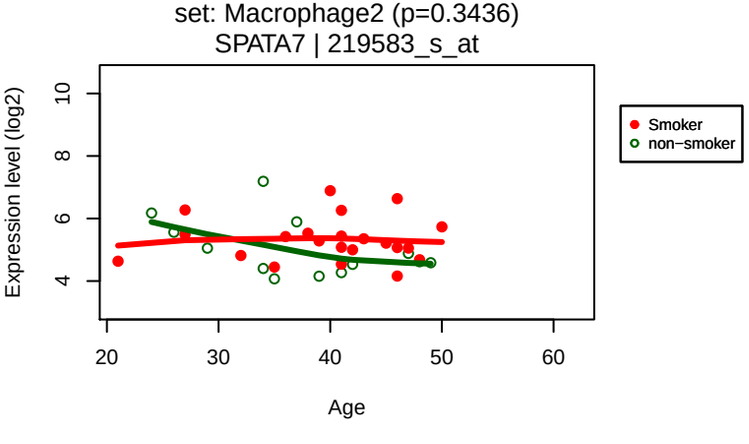

set: SmallAirway2 (p=0.5771)
SPATA7 | 219583_s_at
Expression level (log2)
Age
Smoker
non−smoker

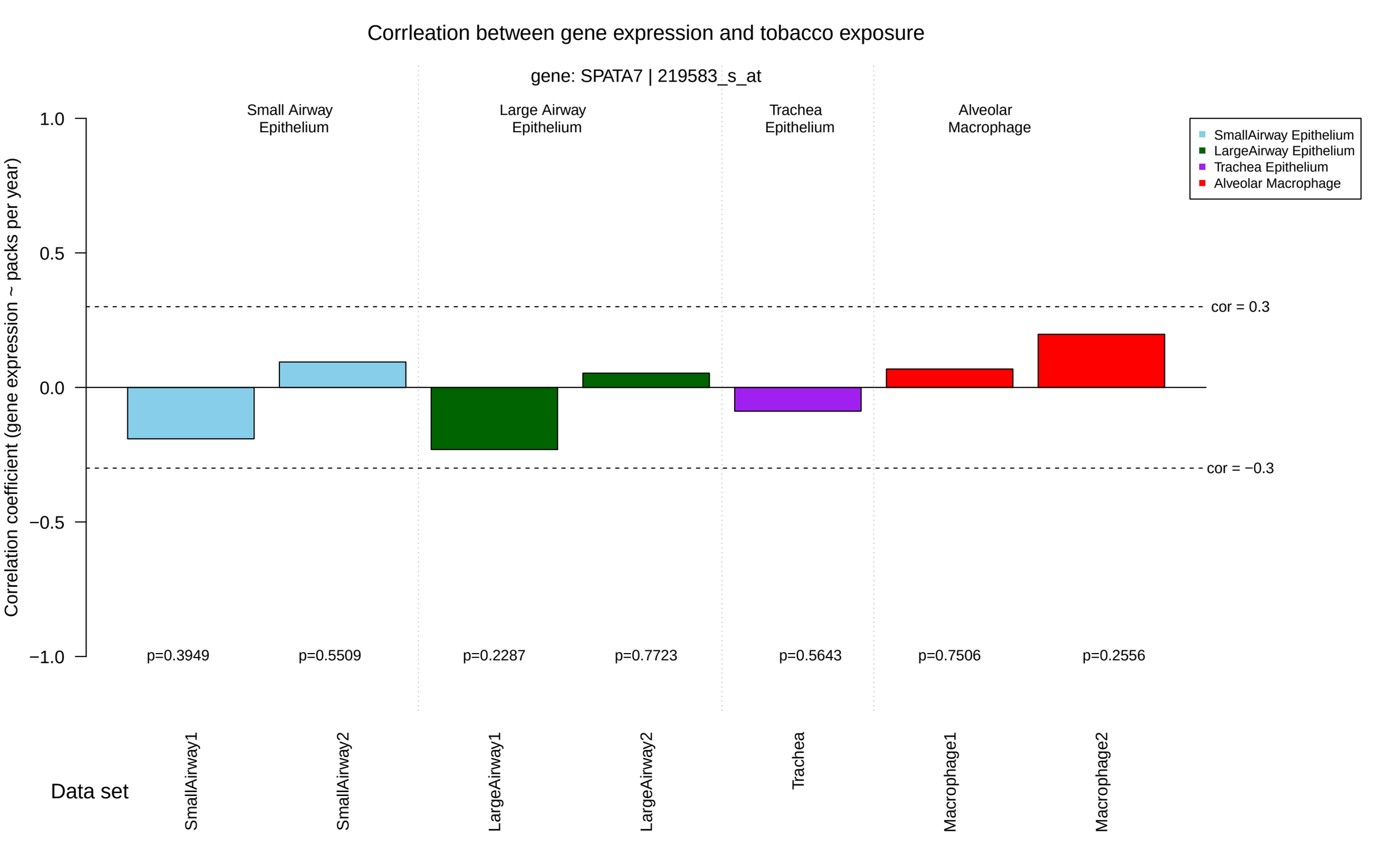

set: LargeAirway1 (p=0.1076)
SPATA7 | 219583_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9643)
SPATA7 | 219583_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.355)
SPATA7 | 219583_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7956)
SPATA7 | 219583_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3436)
SPATA7 | 219583_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SPATA7 | 219583_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3949
p=0.5509
p=0.2287
p=0.7723
p=0.5643
p=0.7506
p=0.2556
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2